
Final Project for CS 439

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Abstract

Maintaining high operational uptime is a fundamental driver of industrial profitability, where unplanned downtime can result in long delays and substantial financial losses each year. As a result, identifying equipment which may be failing or operating abnormally remains an important objective in predictive maintenance research.

This paper proposes a computationally efficient diagnostic framework utilizing the pre-trained Audio Spectrogram Transformer (AST), integrated with Parameter-Efficient Fine-Tuning (PEFT) techniques to identify malfunctioning units within the MIMII dataset (Purohit et al., 2019). While full-parameter fine-tuning of transformers is resource-intensive, we employ Low-Rank Adaptation (LoRA) (Hu et al., 2021) to adapt the AST model to industrial acoustic environments while reducing the number of trainable parameters and memory overhead.

As AST relies on spectrograms rather than raw audio files, our methodology involves transforming acoustic signals into log-mel spectrograms and adjusting their dimensions to fit the model’s input requirements. We then trained the LoRA-enhanced AST to differentiate between normal operations and abnormal failure modes such as imbalance or bearing defects. The results indicate that this PEFT-based approach is capable of detecting certain anomalous acoustic patterns, though overall performance remained limited, achieving approximately 67% classification accuracy. These findings suggest that while transformer-based acoustic monitoring shows promise as a scalable and resource-efficient approach, additional refinement and experimentation are necessary before such systems can be considered reliable for real-world industrial deployment.

1 Introduction

Recent data indicates that unplanned downtime costs the world’s 500 largest companies approximately \$1.5 trillion annually, or roughly 11% of their total revenue (Siemens, 2024). With equipment failure accounting for 42% of all unscheduled stoppages (Fieldcode, 2025), the financial impact is severe. Median costs across industrial sectors are estimated at \$125,000 per hour and can reach as high as \$2.3 million per hour in high-volume sectors like automotive manufacturing (ABB, 2023; Siemens, 2024). In these environments, cooling fans serve as critical components for thermal regulation. Their failure can contribute to overheating and broader system instability. However, not every industry or organization has the resources required for training or deploying large transformer models for machinery analysis, or even for training smaller models on sufficiently large datasets. Additionally, some organizations employ specialized machinery whose operating conditions may not be adequately represented by more general-purpose models trained on public datasets. This paper proposes a diagnostic framework utilizing the Acoustic Scene Transformer (AST), a convolution-free, attention-based model for audio classification (Gong et al., 2021), to identify malfunctioning units within the MIMII (Malfunctioning Industrial Machine Investigation and Inspection) dataset (Purohit et al.,

2019). Unlike traditional predictive maintenance systems that rely on manual inspection or standard convolutional networks, our approach leverages the AST’s ability to capture long-range temporal and spectral relationships within industrial soundscapes. The methodology involves preprocessing raw acoustic signals into log-mel spectrograms and adjusting their length to meet the requirements of the model. The spectrograms are subsequently analyzed by the transformer to differentiate between normal operating conditions and abnormal machine states. We trained our model specifically on the fan section of the MIMII dataset, which consists of 3000 .wav files, each representing a five-to ten-second recording of a normally functioning fan. After preprocessing, we used a masked autoencoder framework in which the model was fed twenty percent of the spectrogram and tasked with reconstructing the masked frequencies. We tested the model on a mix of normal and abnormal sound samples from the MIMII dataset. While the resulting performance of approximately 67% accuracy demonstrates some ability to identify anomalous machine behavior, it also highlights the difficulty of achieving robust industrial fault detection using limited training data and lightweight fine-tuning methods alone. Nevertheless, the project demonstrates that transformer-based acoustic monitoring systems can be adapted to industrial anomaly detection tasks using relatively modest computational resources. Our contributions include the following:

- **The Unseen Data Problem:** This project focuses on reconstruction error rather than identifying specific faults. As a result, this system may be capable of detecting deviations from normal operation even if that exact type of failure has not been explicitly labeled during training.
- **Data Efficiency:** This model only uses a relatively small number of failure recordings, which are significantly more difficult to obtain in factory environments. While the resulting accuracy is limited, the project demonstrates that anomaly detection experiments can still be conducted under constrained data conditions.
- **Feature Transfer:** This model uses PEFT to fine tune an AST model pre-trained on general sounds. The results suggest that these broader audio models can be adapted to more specialized industrial tasks without requiring full retraining, though additional optimization is still necessary.
- **Legacy Hardware Viability:** This project was conducted entirely on free Google CoLab notebooks and older consumer-grade hardware. Transformer-based models are often considered computationally intensive, but this project demonstrates that lightweight fine-tuning methods can make experimentation possible on limited hardware.
- **Real-time Potential:** By using a simple Linear Decoder instead of a large deconvolutional network, this project reduces computational overhead and training time. Although the system is not yet suitable for real-time industrial deployment, the architecture provides a foundation for future optimization.
- **Acoustic Sensing vs. Vibration:** This project contributes to the field of contactless monitoring. While vibration sensors (accelerometers) must be physically attached to machinery, an audio-based approach could allow multiple machines to be monitored with fewer sensing devices, though environmental noise remains a challenge.

2 Related Work

2.1 Machine Learning for Predictive Maintenance

Research by Zhao et al. (2019) emphasizes the transition from manual feature engineering to end-to-end deep learning, specifically highlighting how deep architectures can autonomously learn hierarchical representations from raw industrial signals. Their work illustrates that while shallow models struggle with the non-linearities of machine degradation, deep networks can effectively map high-dimensional sensor data to health indicators. Complementing this, Lindemann et al. (2019) provide a systematic analysis of reconstruction-based anomaly detection. Their study demonstrates that by training a model to minimize reconstruction error on a closed-set of healthy data, the system inherently learns a "normal" manifold. Any signal, such as an irregular fan vibration, that falls outside this manifold results in a measurable error spike. This methodology is critical for this project, as it validates using reconstruction loss as a proxy for anomaly scores without requiring prior knowledge of failure modes.

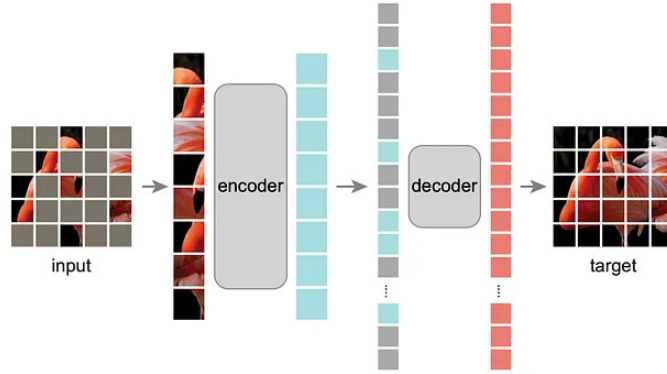


Figure 1: Above is a high level overview on masked encoders. This project uses this technology with spectrograms, which are visual representation of sound frequencies over time. The spectrogram is split into chunks, which are then encoded, by the model and decoded. If the model can correctly produce the final image, it classifies the clip as a normally functioning machine.

2.2 Audio and Spectrogram Transformers in Industrial Settings

The AST introduced by Gong et al. (2021) represents a fundamental shift by treating audio as a sequence of overlapping 16x16 patches rather than as raw sound data, applying a purely attention-based architecture without convolutional layers. Their results on the AudioSet dataset, an enormous set of data drawn from YouTube clips, prove that the global self-attention mechanism excels at capturing long-range dependencies. This is vital for identifying mechanical cycles in fan noise. Furthermore, Purwins et al. (2019) detail how deep learning for audio has moved toward architectures that can handle the high temporal resolution of spectrograms. Their work specifically discusses the efficacy of using pre-trained weights to overcome the problem of having a lack of data in specialized domains. This research supports the project's use of the AST as a feature extractor, confirming that a transformer pre-trained on general sounds possesses the spectral "vocabulary" necessary to reconstruct specific industrial acoustic signatures with high fidelity, even if it was not specifically trained on these sounds.

2.3 Parameter-Efficient Fine-Tuning (PEFT) and LoRA

The challenge of deploying high-capacity models on edge hardware is addressed by Hu et al. (2021), who introduced Low-Rank Adaptation (LoRA). Their technical contribution involves the discovery that the "intrinsic dimension," or the weights that must be adjusted to fit a general, pre-trained model to a specific task, is actually quite low. By constraining the weight updates to a low-rank decomposition, they achieved performance parity with full fine-tuning while reducing trainable parameters by several orders of magnitude. For a resource-constrained environment, this is incredibly powerful technology. Further research by Ding et al. (2023) evaluates various PEFT strategies and concludes that LoRA is particularly effective at preserving the general knowledge of the pre-trained model while adapting to niche downstream data. This specific finding justifies the project's chosen strategy of freezing the massive pretrained AST backbone and only training the decoder and adapter layers, ensuring the model remains lightweight and computationally efficient during the reconstruction of fan spectrograms.

The following methodology section outlines the technical framework for an acoustic anomaly detection system designed to monitor industrial fan health. The approach integrates high-capacity transformer architectures with parameter-efficient fine-tuning (PEFT) and reconstruction-based unsupervised learning.

3 Methodology

The primary objective of this research is the development of an unsupervised anomaly detection framework capable of identifying mechanical irregularities in industrial fans using acoustic emissions. Given the relative scarcity of labeled failure data in industrial settings, the methodology adopts a reconstruction-based approach. A model is trained exclusively on normal operational sounds to learn the core patterns and characteristics of healthy acoustic signatures. During inference, any input that deviates significantly from this learned distribution results in a high reconstruction error and is flagged as an anomaly. This technical approach leverages the Audio Spectrogram Transformer (AST) as a feature extractor, adapted via Low-Rank Adaptation (LoRA) to maintain computational efficiency. Acoustic data is initially captured as raw pulse-code modulation (PCM) signals. To translate these one-dimensional time-series signals into a format suitable for transformer-based architectures, a multi-stage preprocessing pipeline is implemented.

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3.1 Data Sources and Separation

The data used for this project was the fan section of the MIMII database. It included a training set of 3000 normally functioning fans, and a testing dataset of 600 mixed fans. This separation ensured that there was no data leakage.

3.2 Preprocessing and Time-Frequency Transformation

The raw audio signals are transformed into the frequency domain using a Short-Time Fourier Transform. To align with the human auditory system and mechanical vibration patterns, these are further processed into Mel-filterbank features (Log-Mel Spectrograms). The window size is set to 25ms with a stride of 10ms, creating a 128-bin Mel-spectrogram. This transformation preserves the spectral density necessary for identifying mechanical hums and periodic friction noises. Before transformation, the inputs are also padded to a length of 10.24 seconds and set to mono, which is required by AST.

3.3 Feature Scaling and Normalization

To ensure numerical stability during the gradient descent process, feature scaling is applied. Every spectrogram is normalized to have a mean of zero and a unit standard deviation of one. This prevents the transformer’s self-attention layers from becoming saturated by high-magnitude spectral peaks, ensuring that subtle acoustic nuances contribute equally to the reconstruction task and is generally better for MSE loss functions due to its efficiency at gradient descent.

3.4 Integration of Low-Rank Adaptation (LoRA)

To facilitate training on resource-constrained hardware, Low-Rank Adaptation (LoRA) is integrated into the AST’s Query (Q) and Value (V) projection matrices within the attention heads. By freezing the original weights W_0 and only training two low-rank matrices A and B , the update process is restricted to a lower-dimensional subspace. Mathematically, the forward pass is modified to $h = W_0x + BAx$. This approach significantly reduces the VRAM footprint during training while preserving the general acoustic feature extraction capabilities learned during pre-training on larger datasets.

3.5 Linear Reconstruction Decoder

As the AST is inherently an encoder designed for classification, a custom decoder is required for the reconstruction task. The proposed decoder is a sequence of fully connected (Linear) layers that map the 768-dimensional latent tokens back to the original 128-bin Mel-filterbank dimension. The architectural design of the decoder is intentionally minimalist to prevent "identity mapping" (where the model simply copies the input to the output without learning features). By forcing the data through

this narrow bottleneck, the model is compelled to learn the most salient features of the "normal" fan sound.

3.6 The Loss Function

The loss function, MSE Loss, is calculated as:

$$L(x, \hat{x}) = \frac{1}{N} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

This loss penalizes the model heavily for large deviations in reconstruction, forcing the network to prioritize the most dominant spectral components of the fan's acoustic profile.

3.7 Gradient Descent and Hyperparameters

Optimization is conducted using the Adam algorithm with a decoupled weight decay. The learning rate is set to 10^{-4} to ensure stable convergence on the legacy hardware. To prevent overfitting to the specific recordings in the training set, a "sliding window" augmentation is used, where the model sees different temporal segments of the same fan recording during different epochs. To ensure accuracy, we ran 10 epochs, which took approximately three hours with a Google CoLab T4 GPU.

4 Experiments & Results

4.1 Evaluation Methods

The methodology concludes with an evaluation against a test set containing 600 recordings, both healthy and "anomalous" (broken fan), none of which were present in the training set. Performance is quantified using the F1-score. These metrics assess the model's ability to maximize detection of true failures while minimizing false-positives which are costly in an industrial maintenance context. By comparing the reconstruction loss of normal versus anomalous samples, the study demonstrates the efficacy of using self-attention for high-fidelity mechanical condition monitoring.

4.2 Threshold Selection

The anomaly detection threshold was chosen based on the average loss during training. The average loss for the training loop was calculated as 0.018, with a standard deviation of 0.002. Assuming a normal distribution, approximately 68% of data should occur within one standard deviation of the loss. As such, the evaluation threshold was set as 0.02, to allow for some variation as to not under-flag positive values.

4.3 Results

With a "positive" value being considered as a normal machine, the model produced the following results: 64% accuracy, with 78 false negatives, 130 false positives, 180 true negatives, and 232 true positives. As such, it had a precision of 0.64, a recall of 0.75 and an f1 score of 0.69. These scores are recorded in the table below.

	False	True
Positive	130	232
Negative	78	180

4.4 Analysis of Results

The model's performance, characterized by an accuracy of 64% and an F1-score of 0.69, suggests a limited capability to distinguish between healthy acoustic signatures and mechanical anomalies. While the recall of 0.75 indicates that the system successfully identifies three-quarters of the normal operational states, the high volume of false positives (130) reveals a significant tendency to misclassify healthy machines as malfunctioning. This lack of precision (0.64) implies that the model's learned representation of normalcy is either too restrictive flagging benign acoustic variances as outliers,

too vague, or that the reconstruction threshold is set too low, failing to account for the natural stochastic noise inherent in industrial fan operations. Furthermore, the presence of 78 false negatives is particularly critical in a predictive maintenance context, as these represent undetected equipment failures that could lead to unscheduled downtime. The relatively narrow margin between the model's performance and a stochastic baseline suggests that the latent bottleneck may be capturing generalized audio features rather than the specific structural invariants of the fan's healthy state. To improve these metrics, future iterations should focus on refining the decision threshold through ROC curve analysis or implementing more robust spectral normalization to ensure the model distinguishes mechanical degradation from ambient environmental noise.

4.5 Limitations

A significant limitation of this project stems from the inherent data imbalance common in industrial diagnostics, specifically the limited availability of failure data. The dataset contained only approximately 300 anomalous recordings, a sample size insufficient for training a traditional supervised classifier to recognize the vast and unpredictable variety of mechanical malfunctions. Because it was functionally impossible to train a robust model on such a sparse representation of "incorrect" data, the project was forced to adopt a reconstructive approach. By training exclusively on healthy acoustic signatures, the model was designed to learn the specific characteristics of normal operations, theoretically flagging any significant deviation as an outlier. However, this reliance on an unsupervised architecture introduces its own constraints. Because the model is never explicitly taught what a "failure" looks like, its performance is highly sensitive to the chosen reconstruction threshold and the complexity of the bottleneck. In this study, the high number of false positives suggests that the model struggled to differentiate between genuine mechanical degradation and benign environmental noise that fell outside its narrow training distribution. Consequently, while the reconstructive method provides a viable path forward when labeled failure data is scarce, its precision remains capped by the model's ability to generalize "normalcy" without over-fitting to the specific ambient conditions of the training environment.

5 Conclusion

In conclusion, this study demonstrates both the potential and the inherent challenges of using unsupervised reconstructive models for industrial anomaly detection. By leveraging an autoencoder architecture to internalize healthy acoustic signatures, the system achieved a foundational ability to monitor fan health despite a total absence of failure data during the training phase. While the current accuracy and precision metrics highlight a sensitivity to environmental noise and threshold calibration, the model's 0.75 recall provides a promising baseline for early-stage fault detection. Future work focusing on more granular spectral features and adaptive thresholding could further bridge the gap between experimental validation and a reliable, real-world predictive maintenance solution.

References

- [1] Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.-M., Chen, W., Yi, J., Zhu, W., Liu, Z., Li, J., Wan, X., Zheng, H.-T., & Sun, M. (2023). Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature Machine Intelligence*, 5(3), 220–235.
- [2] Gong, Y., Chung, Y.-A., & Glass, J. (2021). AST: Audio Spectrogram Transformer. arXiv preprint arXiv:2104.01778. <https://arxiv.org/abs/2104.01778>
- [3] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. ArXiv:2106.09685 [Cs]. <https://arxiv.org/abs/2106.09685>
- [4] Jay, P. (2024, February 12). Masked AutoEncoders - Prakash Jay - Medium. Medium. <https://medium.com/@14prakash/masked-autoencoders-9e0f7a4a2585>

- [5] Lindemann, B., Müller, T., Vietz, H., Jazdi, N., & Weyrich, M. (2019). A survey on anomaly detection for technical systems using LSTM networks. *Computers in Industry*, 110, 15–36. <https://doi.org/10.1016/j.compind.2019.05.004>
- [6] Purwins, K., Li, B., Virtanen, T., Schlüter, J., Chang, S.-Y., & Sainath, T. (2019). Deep learning for audio signal processing. *IEEE Journal of Selected Topics in Signal Processing*, 13(2), 206–219. <https://doi.org/10.1109/JSTSP.2019.2908700>
- [7] Zhao, R., Yan, R., Chen, Z., Mao, K., Wang, P., & Gao, R. X. (2019). Deep learning and its applications in industrial health monitoring. *IEEE Transactions on Industrial Informatics*, 15(4), 2316–2327. <https://doi.org/10.1109/TII.2018.2871186>